Question 1

1. a) (i)

$$\overline{A}\left(\overline{B+A\overline{C}}\right)$$
$$=\overline{A}\left(\overline{B}\left(\overline{A\overline{C}}\right)\right)$$
$$=\overline{A}\overline{B}\left(\overline{A}+C\right)$$
$$=\overline{A}\overline{B}+\overline{A}\overline{B}C$$
$$=\overline{A}\overline{B}$$

[Remarks to students: Typically, an error can occur on the third line as the bracket term $\overline{A\overline{C}}$ is expanded.]

[4]

ii)

$$B+(C\oplus\overline{B})(AB+\overline{C})$$
$$=B+\left(CB+\overline{C}\overline{B}\right)\left(AB+\overline{C}\right)$$
$$=B+ABC+\overline{C}\overline{B}$$
$$=B+\overline{C}\overline{B}$$
$$=B+\overline{C}$$

[Remarks to students: Careful expansion is required of the XOR bracket term in the first line is required, and of the two bracket terms in the second line.]

[4]

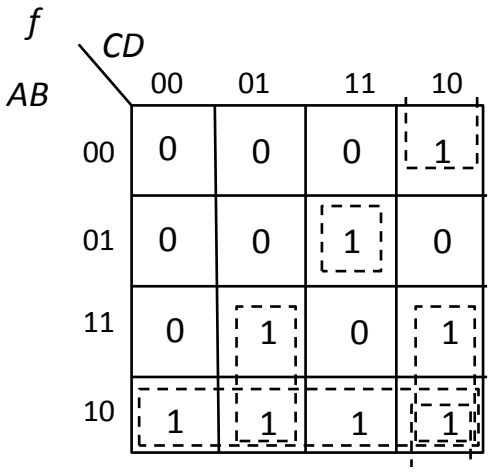
b)

$$f=A\overline{B}+C(\overline{D}\oplus(A\oplus B))+A\overline{C}D$$
$$=A\overline{B}+C(\overline{D}(\overline{A\oplus B})+D(A\oplus B))+A\overline{C}D$$
$$=A\overline{B}+C(\overline{D}\left(AB+\overline{A}\overline{B}\right)+D\left(\overline{A}B+A\overline{B}\right))+A\overline{C}D$$
$$=A\overline{B}+ABC\overline{D}+\overline{A}\overline{B}C\overline{D}+\overline{A}BCD+A\overline{B}CD+A\overline{C}D$$

Here, 2 marks for simplifying the given equation into a form suitable for a Karnaugh map

The equation can now be simplified further using a Karnaugh map:

ANSWERS



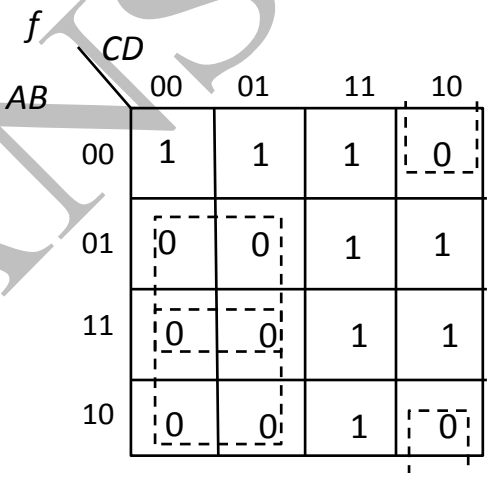
$$\Rightarrow f = A\overline{B} + A\overline{C}D + AC\overline{D} + \overline{B}C\overline{D} + \overline{A}BCD$$

Here, 1 mark for drawing the Karnaugh map, 1 for filling it out correctly, 1 for the correct grouping, and 1 for the final expression.

[Remarks to students: Errors can occur in the expansion of the given Boolean expression into a form such that it can be easily entered into the Karnaugh map.]

[6]

c)



$$\Rightarrow f = \left(\overline{A} + C\right)\left(\overline{B} + C\right)\left(B + \overline{C} + D\right)$$

Here, 1 mark for drawing the Karnaugh map, 1 for filling it out correctly, 1 for the correct grouping, and 1 for the final expression.

ANSWERS

[Remarks to students: Requires you to group zeros and write the result directly in POS form.]

[4]

d)

Decimal	Hexadecimal	Signed binary	Octal
-2049	F7FF		
		0000 0111 1101 0110	3726
4011	FAB		
-14440		1100 0111 1001 1000	

Give 2 marks per answer.

[Remarks to students: Note that the final row has a negative decimal number and the signed binary must be in 2s complement form.]

[8]

e) With two N bit numbers multiplied together, the result has the range:

$$-(2^{2N-2} - 2^{N-1}) \text{ to } 2^{2N-2}$$

[2]

This needs 2N bits. Note that it nearly fits into 2N-1 bits, whose range extends to $2^{2N-2} - 1$.

[2]

[Remarks to students: This requires you to understand the maximum and minimum values possible in a 2s complement system, and hence the number of bits needed to express this range. Note that the most negative number is one greater in magnitude than the most positive number.]

f)

Current state	Input	Next state	Output (PQ)	
S0	0	S0	0	1
S0	1	S1	1	0
S1	0	S2	1	1
S1	1	S1	1	0
S2	0	S0	0	1
S2	1	S1	1	0
S3	X	S0	0	1

Give 1 mark for correct current state, 1 for correct inputs, 1 for correct next state and 1 for correct outputs.

[4]

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[Remarks to students: The circles show within both the state label, and the output.]

g)

A	B	C	S1 = D0	S0 = A	Q
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	1	1	1
1	1	0	0	1	1
1	1	1	1	1	1

Give 2 marks for understanding ROM/multiplexer operation, and 2 each for outputs D0 and Q.

[Remarks to students: This requires you to understand the operation of a ROM, and of a multiplexer. You should be familiar with IEEE notation for both cases.]

[6]

ANSWERS

Question 2

2. (a) (i) The required encoder has 12 input lines I[11:0], where only one of these can be zero (active) at a time. There are four output lines D[3:0], giving the binary equivalent of the hexadecimal values 0 – B.

2 marks for understanding the above.

The truth table is then as follows:

Number	I11	I10	I9	I8	I7	I6	I5	I4	I3	I2	I1	I0	D3	D2	D1	D0
0	0												0	0	0	0
1		0											0	0	0	1
2			0										0	0	1	0
3				0									0	0	1	1
4					0								0	1	0	0
5						0							0	1	0	1
6							0						0	1	1	0
7								0					0	1	1	1
8									0				1	0	0	0
9										0			1	0	0	1
10 (A)											0		1	0	1	0
11 (B)												0	1	0	1	1

For clarity, the deactivated input values in the above table are left as empty cells. These are occupied by 1s (deactivated).

2 marks for the inputs and 2 for the outputs.

[6]

[Remarks to students: Here, errors can occur in understanding (i) what an encoder is, (ii) the active low inputs and active high outputs, and the number of row, I0 – I11.]

(ii) The Boolean expressions for D[3:0] can be seen by inspection:

$$D0 = \left(\overline{I1}\right) + \left(\overline{I3}\right) + \left(\overline{I5}\right) + \left(\overline{I7}\right) + \left(\overline{I9}\right) + \left(\overline{I11}\right)$$

$$D1 = \left(\overline{I2}\right) + \left(\overline{I3}\right) + \left(\overline{I6}\right) + \left(\overline{I7}\right) + \left(\overline{I10}\right) + \left(\overline{I11}\right)$$

$$D2 = \left(\overline{I4}\right) + \left(\overline{I5}\right) + \left(\overline{I6}\right) + \left(\overline{I7}\right)$$

$$D3 = \left(\overline{I8}\right) + \left(\overline{I9}\right) + \left(\overline{I10}\right) + \left(\overline{I11}\right)$$

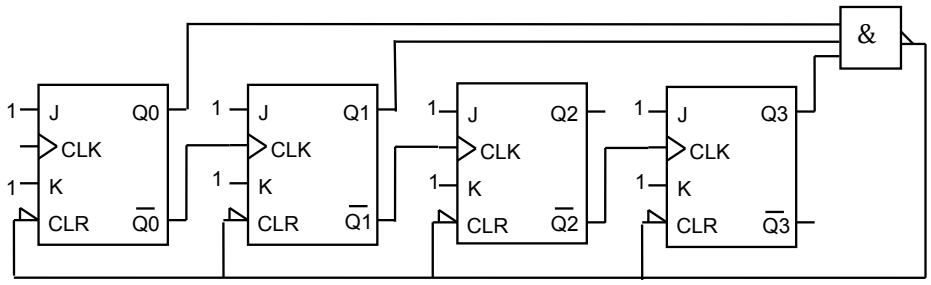
2 marks for each expression.

[8]

[Remarks to students: Given that only one input at a time is active, the Boolean expressions can be written directly. Note the bars, as the inputs are active low.]

ANSWERS

(b)



4 marks for correct use of the J-K flip-flops and 2 for the asynchronous reset.

[Remarks to students: This requires an asynchronous binary counter, where the clock is cascaded from one flip-flop to the next. The reset occurs on 1011 (B), to allow 1010 (A) to persist for one clock cycle.]

[6]

c) (i) The given Boolean function is:

$$f = \overline{A}\overline{B} + AB + \overline{A}C$$

This corresponds to the truth table:

A	*B*	*C*	*f*
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

[1]

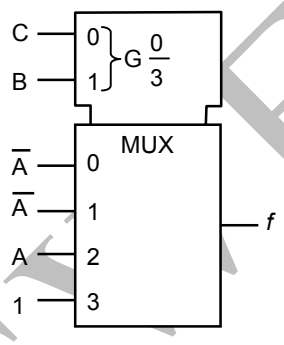
As BC must be used on the select lines of the 4 × 1 MUX, we draw a k-map table, with BC arranged along the rows and A along the columns. Furthermore, we choose to connect C = S0 and B = S1.

ANSWERS

A \ S1 S0 (BC)	00	01	11	10
0	1	1	1	0
1	0	0	1	1
	$I0 = \overline{A}$	$I1 = \overline{A}$	$I3 = 1$	$I2 = A$

[2]

This gives the following implementation:



Note: The alternative solution has B = S0 and C = S1. We would then have I0 = A(bar), I1 = A, I3 = 1, and I2 = A(bar).

[1]

[Remarks to students: You need to identify how the variables may be connected to the input lines of the multiplexer]

(ii) Using two 2 × 1 MUXs, we first need to rearrange the given function in the form of the Boolean function for the output Z for a 2 × 1 MUX, with select input S:

$$Z = \overline{S}I0 + SI1$$

For the given function, we then have:

$$f = \overline{A}\overline{B} + AB + \overline{A}C$$
$$= \overline{A}\left(\overline{B} + C\right) + AB$$

[2]

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[Remarks to students: Arranging the given expression into the form of the equation for a 2 × 1 multiplexer can cause problems.]

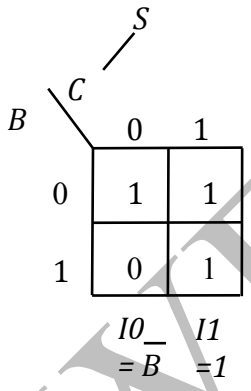
Comparison with the expression for Z implies that:

$$I0 = \left(\overline{B} + C\right) = g$$

$$I1 = B$$

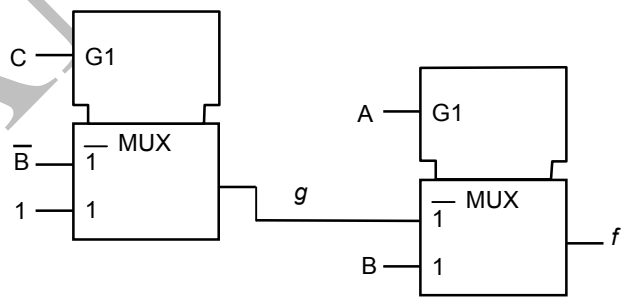
$$S = A$$

We then need to implement $\overline{B} + C$ using a second 2 × 1 MUX. Here, either B or C could be connected to the select line. Choosing C for the select line (the alternative of B on the select line is also fine) and inspecting the following map:



[2]

This allows implementation of the second 2 × 1 MUX, with I0 = B(bar) and I1 = 1. We then have the following final implementation (other solutions which work are also possible, full marks should be given for any working solution using the two multiplexers and if needed, some inverters):



[2]

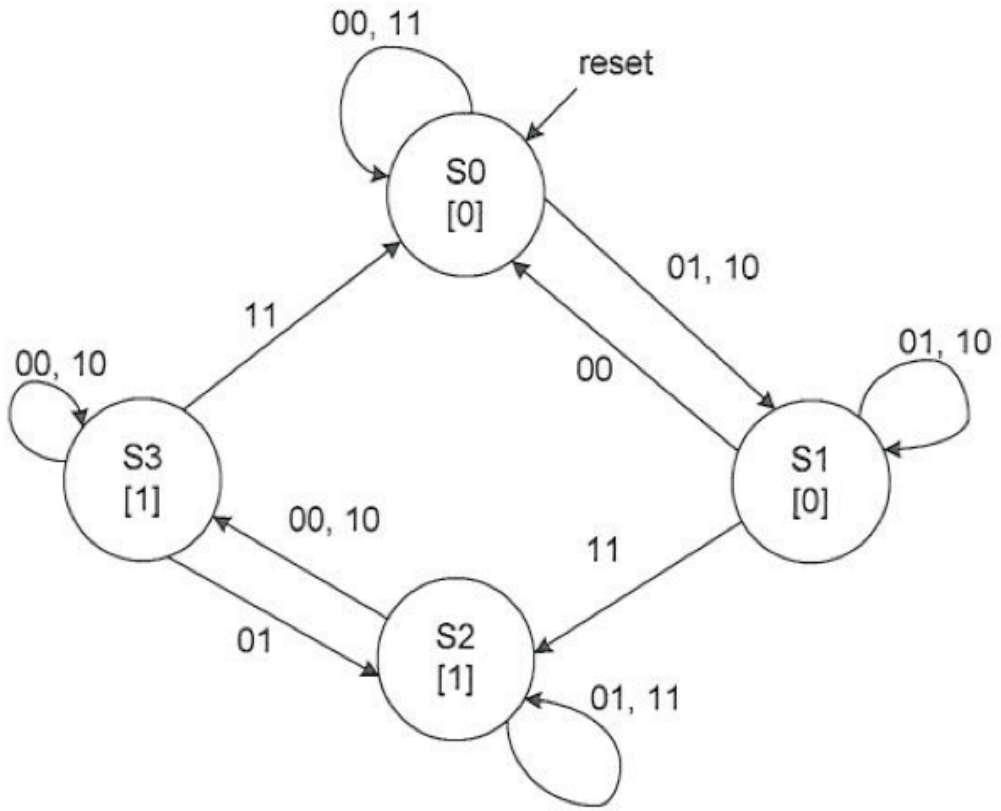
[Remarks to students: Various solutions are possible, as the original expression can be arranged in multiplexer form in various ways. The above explains the approach for one of these.]

ANSWERS

Question 3

3. (a) Using the states S0, S1, S2 and S3 to define the FSM, the following state diagram can be obtained. Note that as there are two inputs X1, X2, each state has $2^2 = 4$ transitions possible. Furthermore, a sequence of two input combinations, one after the other, is needed to change the output. For example, for the sequence X1X2 = 01, 11, the first part, X1X2 = 01 causes a transition from S0 to S1, and the second part, X1X2 = 11, causes a transition from S1 to S2, and Z = 0 to Z = 1.

Give 2 marks for identifying this reasoning.



Give 2 marks for showing four states, 4 for the correct interconnections, and 4 for the correct input/output values.

[12]

[Remarks to students: Drawing this diagram proved difficult for many students. Note that for each state four possible outcomes exist, as there are four input combinations, 00, 01, 10, and 11, in each case. The output changes between 0 and 1 only after a resting state, S1 or S3, has been passed. The answer required working carefully through the definition of this FSM, as given in the question.]

(b) Using the encoding system given, the state diagram can be converted into the following table, with Q3Q2Q1Q0 as the present state of the FSM, and $Q3^+Q2^+Q1^+Q0^+$ as the next state:

Q3	Q2	Q1	Q0	X1	X2	$Q3^+$	$Q2^+$	$Q1^+$	$Q0^+$	Z
0	0	0	1	0	0	0	0	0	1	0
0	0	0	1	0	1	0	0	1	0	0
0	0	0	1	1	0	0	0	1	0	0
0	0	0	1	1	1	0	0	0	1	0
0	0	1	0	0	0	0	0	0	1	0
0	0	1	0	0	1	0	0	1	0	0
0	0	1	0	1	0	0	0	1	0	0
0	0	1	0	1	1	0	1	0	0	0
0	1	0	0	0	0	1	0	0	0	1
0	1	0	0	0	1	0	1	0	0	1
0	1	0	0	1	0	1	0	0	0	1
0	1	0	0	1	1	0	1	0	0	1
1	0	0	0	0	0	1	0	0	0	1
1	0	0	0	0	1	0	1	0	0	1
1	0	0	0	1	0	1	0	0	0	1
1	0	0	0	1	1	0	0	0	1	1

Give 0.5 marks for each row.

[8]

[Remarks to students: Once the state diagram has been drawn, writing the above table should be straightforward.]

(c) As the FSM states are encoded using 1-hot encoding, the Boolean equations can be written directly by inspection:

$$Q0^+ = Q0\overline{X1}\,\overline{X2} + Q0X1X2 + Q1\overline{X1}\,\overline{X2} + Q3X1X2$$

$$Q1^+ = Q0X1\overline{X2} + Q0\overline{X1}X2 + Q1\overline{X1}X2 + Q1X1\overline{X2}$$

$$Q2^+ = Q1X1X2 + Q2\overline{X1}X2 + Q2X1X2 + Q3\overline{X1}X2$$
$$= Q1X1X2 + Q2X2 + Q3\overline{X1}X2$$

$$Q3^+ = Q2\overline{X1}\,\overline{X2} + Q2X1\overline{X2} + Q3\overline{X1}\,\overline{X2} + Q3X1\overline{X2}$$
$$= Q2\overline{X2} + Q3\overline{X2}$$

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$Z = Q2 + Q3$

Give 2 marks for each Boolean expression.

[Remarks to students: The above expressions can be written by inspection, as there is a '1' for only one of the four bits defining a given state in 1-hot encoding. A Karnaugh map approach is impractical and unnecessary.]

[10]

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